

Beginner Level

Use Case 1: Text Completion Bot with LangChain

Requirements:

- Use LangChain to connect to a pre-trained language model (e.g., OpenAI's GPT-4) to complete sentences or paragraphs.
- Implement a simple UI using Streamlit for input and output.
- Basic error handling and validation for input text.

Evaluation Criteria:

- Correct implementation of LangChain API calls.
- User-friendly interface for input and output.
- Basic error handling implemented.
- Code quality and readability.

Intermediate Level

Use Case 2: Chatbot for FAQ using Semantic Kernel AI Agents

Requirements:

- Develop a chatbot that answers frequently asked questions using Semantic Kernel AI Agents.
- Load a set of FAQs and answers from a file or database.
- Match user queries with the closest FAQ and provide a response using Semantic Kernel's retrieval-augmented generation capabilities.

Evaluation Criteria:

- Accuracy of matching user queries to FAQs.
- Ability to handle various phrasings of the same question.
- Performance and response time.
- Robustness and error handling.

Intermediate Level

Use Case 3: Image Captioning with LangChain and OpenAI API

Requirements:

- Use LangChain to connect to a pre-trained image captioning model (e.g., OpenAI's DALL-E) to generate descriptions for images.

- Develop a simple web application using Flask or FastAPI to upload images and display captions.
- Handle different image formats and sizes.

Evaluation Criteria:

- Correct implementation of the image captioning model.
- User-friendly image upload and caption display interface.
- Handling of various image formats and sizes.
- Quality of generated captions.

Advanced Level

Use Case 4: Code Generation Assistant with Semantic Kernel AI Agents

Requirements:

- Develop a tool that generates code snippets based on natural language descriptions using Semantic Kernel AI Agents.
- Implement a simple IDE-like interface for users to input descriptions and view generated code.
- Fine-tune the model with code-specific data to improve accuracy.

Evaluation Criteria:

- Accuracy and relevance of generated code snippets.
- User interface and user experience.
- Error handling and suggestions for improvement.
- Integration with a version control system (optional).

Advanced Level

Use Case 5: Personalized Content Generator with LangChain and Agent Framework

Requirements:

- Create a system that generates personalized content (e.g., blog posts, marketing emails) based on user profiles and preferences using LangChain and Agent Framework.
- Implement user authentication and profile management.
- Fine-tune the language model with specific data for personalization.

Evaluation Criteria:

- Personalization accuracy based on user profiles.
- Quality and relevance of generated content.
- Security and privacy of user data.

- Scalability and performance of the system.

General Evaluation Criteria

Technical Criteria:

- Correct implementation of the generative AI models.
- Effective use of LangChain, Semantic Kernel AI Agents, and Agent Framework.
- Performance and scalability of the solution.
- Error handling and robustness of the application.

Role	Code Readability	Concept Understanding	Innovation	Research	Presentation	Handling of errors
Developer						
DevOps						
Tester						
Implementation						

Non-Technical Criteria:

- User interface and user experience.
- Creativity and innovation in solving the use case.
- Documentation and code quality.
- Presentation and explanation of the solution.

These use cases provide a structured approach to training and evaluating developers on Generative AI, leveraging the latest frameworks and tools.